Dual-Branch Domain Adaptation Few-Shot Learning for Hyperspectral Image Classification

Zhuowei Wang, Shihui Zhao, Genping Zhao, and Xiaoyu Song

***Abstract*— Cross-domain few-shot learning (FSL) often employs adversarial domain adaptation techniques to address the issue of data distribution discrepancies between the source and target domains. However, forcing the alignment of two distinct domains may lead to distortions in class distribution alignment and result in a decrease in classification performance in hyperspectral image (HSI) analysis. Moreover, existing cross-domain methods are often applied to satellite/airborne HSIs as both the source and target domains. It is rarely explored whether the same cross-domain methods can be applied for cross-applications where the source and target domain data could be both satellite/airborne HSIs with lower spatial resolution and unmanned aerial vehicle (UAV) HSIs with higher spatial resolution. To address these issues, this article proposes a novel domain-adaptive FSL network with dual branches, respectively, aiming at domain fusion and domain separation. The domain fusion branch uses a conditional adversarial network to align the global distributions of the two domains, while the domain separation branch introduces a gate mechanism for discriminative feature learning in each domain to achieve independent category distributions. During the experiment, the proposed method is evaluated by performing cross-transfer learning under the condition that low-spatial-resolution hyperspectral data and high-spatial-resolution hyperspectral data are used as source and target data alternately. The experimental results suggest that the proposed method not only mitigates the negative effects of forced alignment in domain fusion, but also holds potential for cross-domain transfer learning between low- and high-spatial-resolution HSIs.**

***Index Terms*— Domain adaptation, few-shot learning (FSL), gate mechanism, hyperspectral image (HSI) classification.**

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Zhuowei Wang and Shihui Zhao are with the School of Computer Science and Technology, Guangdong University of Technology, Guangzhou 510006, China (e-mail: wangzhuowei0710@163.com; 907126738@qq.com).

Genping Zhao is with the School of Computer Science and Technology, Guangdong University of Technology, Guangzhou 510006, China, and also with Beijing Piesat Information Technology Company Ltd., Beijing 100095, China (e-mail: genping.zhao@gdut.edu.cn).

Xiaoyu Song is with the Department of Electrical and Computer Engineering, Portland State University, Portland, OR 97207 USA (e-mail: songx@pdx.edu).

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I. Introduction

HYPERSPECTRAL images (HSIs) consist of hundreds to thousands of spectral bands, containing rich spatial–spectral information [1]. Its distinctive attributes propel hyperspectral imaging into extensive application across domains including environmental studies [2], [3] and military operations, encompassing tasks such as detection, prediction, and reconnaissance [4], [5]. Therefore, HSI classification has become a highly focused research area [6], [7], [8]. Known for the powerful feature learning capabilities of deep learning in various domains, the classification of hyperspectral remote-sensing images through the utilization of deep-learning methods has emerged as more preferred approach in recent years [9], [10], [11], [12], [13]. However, HSIs usually have unlabeled or sparsely labeled samples, while deep networks reliant on data generally demand a substantial amount of training information. Therefore, the prominent issue of small sample size in hyperspectral remote-sensing images becomes a crucial factor limiting the further enhancement of deep learning for HSI classification performance. Simultaneously, labeling HSIs is a time-consuming, labor-intensive, and expensive process. Consequently, addressing the restriction of labeled samples poses a pivotal obstacle in HSI classification.

Previous research has explored effective feature learning methods for unlabeled HSI learning using unsupervised encoding networks. For instance, Mei et al. [14] presented a technique for spatial–spectral feature learning strategy employing an unsupervised 3-D convolutional autoencoder (3D-CAE). Xu et al. [15] introduced the spectral–spatial semantic feature learning network (S3FN) to achieve unsupervised deep extraction of semantic features from HSIs. For HSIs with limited labeled samples, semisupervised learning approaches can be utilized. As an illustration, Fan et al. [16] proposed a dataset generation approach for unlabeled samples by employing a semisupervised multihead convolutional neural network (CNN) technique with ensemble learning methods. Jia et al. [17] introduced a semisupervised autoencoder Siamese network (3DAES) that leverages unsupervised feature learning from unlabeled data using autoencoders and utilizes a limited labeled sample set for calibration. Additionally, an approach combined with active learning is also employed. Li et al. [18] developed an active learning prototype network (ALPN) that extracts distinctive features from a limited number of samples. This network combines semisupervised and active learning techniques to increase the number of annotated

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samples. However, the aforementioned neural network-based learning methods that incorporate unsupervised, semisupervised, and active learning approaches mainly focus on the analysis of the data for small-sample HSIs in the target domain. They did not make use of existing HSI data with sufficient labels as source data to guide target data learning.

With respect to the fact of limited labeled samples, cross-domain HSI classification tasks through transfer learning make it possible to utilize publicly available HSI data with sufficient labels. Domain adaptation methods primarily modify the feature space of the source and target domains to minimize the distribution discrepancy between them [19], [20], [21]. For example, Huang et al. [22] proposed a spatial–spectral weighted adversarial domain adaptation (SSWADA) network for cross-scene wetland mapping using HSIs, where a weighted adversarial discrimination strategy is employed to align the feature distributions between source and target scenes. Zhang et al. [23] developed a topology structure and semantic information transmission network (TSTnet), in which the distribution alignment between the source domain and the target domain is achieved through maximum mean discrepancy (MMD). Zhang et al. [24] developed a single-source domain expansion network (SDEnet), in which a generator, comprising a semantic encoder and morph encoder, and a discriminator, incorporating supervised contrastive learning, are employed to learn domain-invariant information through adversarial learning. The domain distribution alignment in these domain adaptation methods is essentially aimed at achieving knowledge transfer for sharing information. However, domain adaptation relies on the presumption that the two domains exhibit similarities in terms of categories. This prevents them from efficiently categorizing classes in the test target data that have not been trained before. To overcome this limitation, it is expected that a model can learn to learn so that to address these issues. Meta-learning [25] is a type of model inspired by human cognitive systems that possess the prowess of "learn to learn." Meta-learning models can generalize available knowledge to new data. Few-shot learning (FSL) is a specific application of meta-learning [26]. This involves transferring meta-knowledge from a few labeled instances to unknown data. To obtain FSL-compatible meta-knowledge, FSL simulates the few-shot task by partitioning the data into episodes and performing training on multiple identical episodes. With the meta-knowledge-equipped model, few-shot HSI classification can be achieved. For example, Gao et al. [27] utilized relationship networks imbued with the concept of meta-learning to devise a deep classification model suitable for few-shot tasks.

The premise for effective classification in most FSL approaches is that the source and target domains have the same data distribution. However, in practical remote-sensing applications, HSIs acquired from different sensors exhibit variations in spectral range, spatial resolution, and categories. Domain alignment via cross-domain FSL network is the common strategy to address these problems. For example, Li et al. [28] introduced a deep cross-domain FSL (DCFSL) network that combines domain adaptation and FSL. Zhang et al. [29] developed a framework called graph information aggregation cross-domain FSL (Gia-CFSL) that integrates FSL with graph-based domain alignment. The transformer is also introduced into domain adaptation FSL to fully learn the global features of HSIs of both the source and target domains [30]. Peng et al. [31] proposed an FSL approach based on a convolutional transformer (CT) network (CTFSL), in which the CT extracts both local and global information, and an adversarial network is employed for domain alignment. Despite the improvements in few-shot HSI classification performance achieved by these methods, they do not take into account the potential negative impact of aligned data distributions. In fact, domain alignment forcibly aligns the category distributions of two domains leading to distortions in category distributions. This would affect the distinguishability between categories, especially when dealing with datasets that have extremely different data sources, and subsequently reducing the performance of HSI classification.

As with the development of unmanned aerial vehicle (UAV) imaging, there are increasing applications with UAV-based hyperspectral imaging which is of high spatial resolution. It is more flexible to conduct image labeling for UAV HSIs than satellite/airborne images of low spatial resolution. Knowledge transfer conducted between HSIs with low and high spatial resolutions is faced with extremely distinct data distribution. Therefore, it is essential to have the cross-domain learning network maintain category independence when aligning data distributions of two domains in domain adaptation. However, rarely do studies spend effort on learning the discriminative knowledge of both domains when conducting domain alignment. Moreover, available studies undertake more investigations on cross-domain FSL between public satellite/airborne HSIs which have low spatial resolution. In fact, it is of great significance to figure out the different performance of cross-domain FSL when transfer learning is undertaken between low- and high-spatial-resolution images alternatively.

To address the aforementioned problem, a dual-branch domain adaptation FSL network is proposed in this study to achieve HSI classification. The main contributions of this article are threefold.

1) A dual-branch domain adaptation FSL method based on domain separation and domain fusion is proposed to achieve domain data distribution alignment while maintaining domain class distribution independence. Compared to the available DCFSL network using domain fusion learning to align the distribution of the two domains, the domain separation learning branch is added by developing a domain gate (DG) mechanism network structure to learn discriminative knowledge.
2) Investigation of cross-domain transfer learning using HSIs with low spatial resolution and HSIs with high spatial resolution alternatively as the source dataset and the target dataset is conducted to evaluate the different performance when the spatial resolutions of the source and target domain data are changed.
3) The experimental results demonstrate that the proposed dual-branch domain adaptation approach effectively addresses the negative impact caused by domain adaptation. It outperforms available state-of-the-art

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counterparts of CNN-associated networks, related networks of DCFSL, and even the latest advanced CTFSL with respect to classification accuracy. Of more importance, its superiority over the latest CTFSL is with respect to both model efficiency and precise classification. In addition, the proposed model presents potential to be used for cross-domain transfer tasks between low- and high-spatial-resolution HSIs. It shows especially dominant merit compared to CTFSL when learning from high-spatial-resolution HSIs to transfer on low-spatial-resolution HSIs.

II. Methodology

Assume that the source-domain hyperspectral dataset $\boldsymbol{D}_S$ has C_S classes and the target domain dataset $\boldsymbol{D}_T$ has C_T classes. In FSL, the training set consists of $\boldsymbol{D}_S$ and a small labeled target dataset $\boldsymbol{D}_{\text{tar}}$. Subsequently, the trained model is evaluated using the unlabeled target dataset $\boldsymbol{D}_{\text{tes}}$. It is important to note that $\boldsymbol{D}_{\text{tar}} \cup \boldsymbol{D}_{\text{tes}} = \boldsymbol{D}_T$, $\boldsymbol{D}_{\text{tar}} \cap \boldsymbol{D}_{\text{tes}} = \phi$, and $C_S \cap C_T = \phi$. To obtain meta-knowledge for FSL, in this study, the few-shot task is conducted by making use of meta-learning which needs to divide $\boldsymbol{D}_S$ and $\boldsymbol{D}_{\text{tar}}$ into episodes. Each episode consists of an N-way-K-shot setup, where N refers to the number of classes randomly selected from the training data and K represents the number of samples per class within the support set, denoted as $\boldsymbol{S} = \{(\boldsymbol{x}_i, y_i)\}_{(i=1)}^{(N\times K)}$. Additionally, the query set, indicated as $\boldsymbol{Q} = \{(\boldsymbol{x}_j, y_j)\}_{(j=1)}^{(N\times W)}$, contains W samples per class. The episodes, $\boldsymbol{F}_s = \{\boldsymbol{S}_s, \boldsymbol{Q}_s\}$, and $\boldsymbol{F}_t = \{\boldsymbol{S}_t, \boldsymbol{Q}_t\}$, are sampled from $\boldsymbol{D}_S$ and $\boldsymbol{D}_{\text{tar}}$, respectively. Here, $\boldsymbol{x}$ represents a pixel block and y represents the corresponding label for the pixel. Considering the involvement of a handful of labeled instances from the target data in training, in this study, to address the sample bias issue and match the data sizes between the target and source domains, data augmentation using the Gaussian noise method [32] is performed at the data preparation stage. Specifically, new HSI $\boldsymbol{x}'$ is generated through multiplying certain training sample $\boldsymbol{x}$ from the target domain by a random factor α and adding Gaussian noise $\boldsymbol{noi}$ as follows:

$$\boldsymbol{x}' = \alpha \boldsymbol{x} + \beta \boldsymbol{noi} \tag{1}$$

where α is randomly sampled from a uniform distribution with a range of (0.9, 1.1), $\beta = 0.04$. Gaussian noise $\boldsymbol{noi}$ is drawn randomly from a normal distribution with a mean of 0 and a standard deviation of 1. As only additive and multiplicative transformations are operated, $\boldsymbol{x}'$ has the same size as $\boldsymbol{x}$.

A. Overall Framework

The schematic of the proposed dual-branch domain adaptation training network framework is shown in Fig. 1. As seen from the figure, random sampling is first conducted to divide the available training data into episodes. Following this, those training samples go through 2-D convolutions to unify the multimodality data into the same feature space. Subsequently, shallow spatial–spectral feature learning is performed through a 3-D residual network structure. The features are then fed into two dual-branch networks, namely domain fusion and domain separation branches, to learn different information features. Finally, the two losses generated from the domain separation branch and domain fusion branch are combined to form the overall network loss function. This loss function is applied for model training and obtaining a transferable classification model, which can be used to classify unlabeled target data.

B. Shallow Learning Structure Module

Considering the spectral dimension mismatch between the two domains, an initial preprocessing is performed on datasets in both domains using 2D-CNN mapping units to reduce the dimensionality. This preprocessing step reduces the original feature dimensions of the image cubes of two domains to a feature dimension of 100, resulting in HSI cube $\boldsymbol{x} \in R^{100\times 9\times 9}$ [33]. After unifying the feature dimension, the 3-D residual CNN feature extraction module E_1 is employed to conduct shallow learning on data from both domains. This model is composed of one residual block and one max-pooling layer, which renders a summary of preliminary semantic information. Among them, the 3-D residual block [28] consists of three 3-D convolutional layers, each with a kernel size of $3 \times 3 \times 3$ and a total of eight convolutional kernels, as shown in Fig. 2. Through this stage of shallow learning, the abstract features such as texture, edges, and corners of input images can be initially captured. They are common representations for image data of both the source and target domains.

C. Dual-Branch Domain Adaptation Feature Learning Deep Network

To address feature learning in cross-domain data, it is necessary to simultaneously consider the shared invariant features and distinguishable differential features of the source and target domains. To achieve this goal, this article introduces a dual-branch domain adaptation network framework that combines domain separation and domain fusion for FSL. Namely, after universal feature representation through shallow learning, further higher-level semantic features are learned using a domain-fused and domain-separated dual-branch network.

1) Domain Fusion Branching Network: In the available FSL network of DCFSL [28], both domain datasets are concurrently utilized for model training, signifying that the model learns the common characteristics inherent in both domains. Therefore, as illustrated in Fig. 1, the domain fusion branch network in this study adopts the same structure as DCFSL [28]. It includes a 3-D convolution residual network E_2 to accomplish the shared spatial–spectral feature learning of mixed data. The difference between the 3-D residual block in E_2 and the shallow learning module is that the number of convolution kernels is 16. Subsequently, prototype learning [34] is applied to learn prototypes of the generalized spatial–spectral feature data. To elucidate with an example from the source data, let f_ω represent the complete learning function of the shallow network E_1 and the deep network E_2. After learning through the shallow network and the deep network, the output feature representation of the support set is denoted as $f_\omega(\boldsymbol{x}_i)$. The prototype $\boldsymbol{p}_s$ for each class is the mean vector of $f_\omega(\boldsymbol{x}_i)$,

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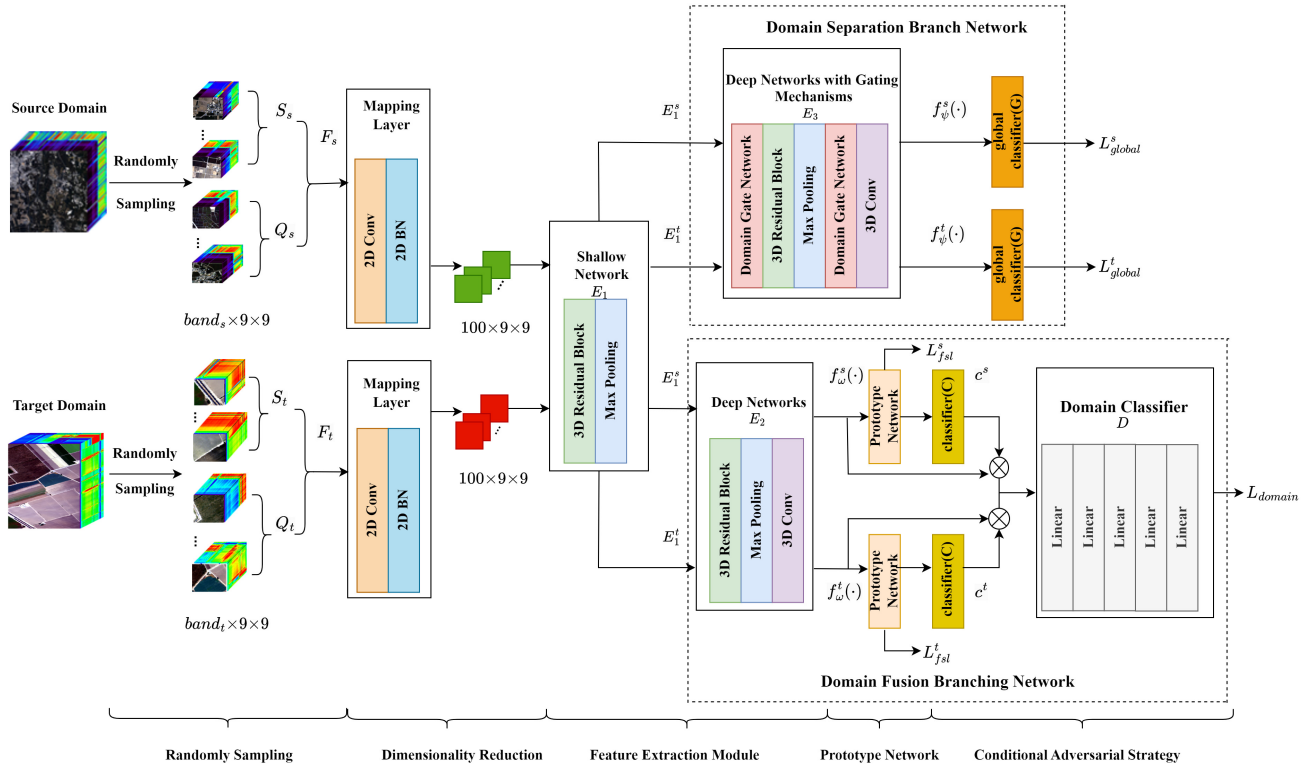

Fig. 1. Training framework of the proposed dual-branch domain adaptation FSL network. This framework includes preprocessing units for the input datasets to be resized into small patches of unified dimensions, shallow network structure to conduct early period feature learning, and the main body of dual-branch network structure which includes a domain fusion branch to learn aligned common features and a domain separation branch where a DG network is contained to use a gating mechanism to achieve domain separation.

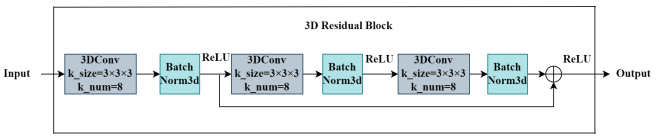

Fig. 2. Detailed description of the 3-D residual block.

which is given by the following equation:

$$\boldsymbol{p}_s = \frac{1}{|\boldsymbol{S}_s|} \sum_{(\boldsymbol{x}_i, y_i) \in \boldsymbol{S}_s} f_\omega(\boldsymbol{x}_i). \tag{2}$$

Therefore, in each episode of the query set, the prediction probability P_ω for the query sample $\boldsymbol{x}_j$ is calculated as follows based on prototype learning:

$$P_\omega\left(y_j = k | \boldsymbol{x}_j \in \boldsymbol{Q}_s\right) = \frac{\exp\left(-d\left(f_\omega\left(\boldsymbol{x}_j\right), \boldsymbol{p}_s\right)\right)}{\sum_{k=1}^{N} \exp\left(-d\left(f_\omega\left(\boldsymbol{x}_j\right), \boldsymbol{p}_s\right)\right)}. \tag{3}$$

The source-domain FSL loss L^s_{fsl} for each episode is expressed as the negative logarithm of the probability based on $\boldsymbol{x}_j$ and its true class label k, as shown in the following equation:

$$L^s_{\text{fsl}} = E_{\boldsymbol{S}_s, \boldsymbol{Q}_s}\left[- \sum_{(\boldsymbol{x}, y) \in \boldsymbol{Q}_s} \log P_\omega(y = k \mid \boldsymbol{x}) \right]. \tag{4}$$

Similar to the source domain, the FSL loss L^t_{fsl} for each episode in the target domain is calculated by the following equation:

$$L^t_{\text{fsl}} = E_{\boldsymbol{S}_t, \boldsymbol{Q}_t}\left[- \sum_{(\boldsymbol{x}', y) \in \boldsymbol{Q}_t} \log P_\omega\left(y = k \mid \boldsymbol{x}'\right) \right]. \tag{5}$$

After prototype learning, a basic linear classifier C is employed to categorize the prototype representation features of each domain to get discriminative classification results in $\boldsymbol{c}$. The predicted classification result $\boldsymbol{c}$ from classifier C and the learned feature $\boldsymbol{f} = f_\omega(\boldsymbol{x})$ are then linearly mapped (realized by operation $\otimes$) and fed into a discriminator D, which consists of five fully connected layers. In this way, the usage of features $\boldsymbol{f}$ is aligned, forming the conditional domain adversarial network (CDAN) structure [35].

The linear mapping $\otimes$ is capable of thoroughly capturing the multimodal structure that emerges after complex data distributions are adequately apprehended. However, its drawback lies in its susceptibility to dimension explosion. Therefore, the following random sampling strategy is adopted to tackle this issue:

$$T(\boldsymbol{f}, \boldsymbol{c}) = \begin{cases} T_\otimes(\boldsymbol{f}, \boldsymbol{c}), & d_f \times d_c \leq 1024 \\ T_\odot(\boldsymbol{f}, \boldsymbol{c}), & \text{otherwise} \end{cases} \tag{6}$$

where d_f and d_c represent the dimensions of $\boldsymbol{f}$ and $\boldsymbol{c}$, respectively, the dimension of the linear mapping $\boldsymbol{f} \otimes \boldsymbol{c}$ is given by $d_f \times d_c$. A multilinear mapping is performed by randomly selecting certain dimensions of $\boldsymbol{f}$ and $\boldsymbol{c}$: $T_\odot(\boldsymbol{f}, \boldsymbol{c}) = (1/|(d_s)^{1/2}|)(\boldsymbol{R}_f \boldsymbol{f}) \odot (\boldsymbol{R}_c \boldsymbol{c})$, where $\odot$ denotes element-wise

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operation and $\boldsymbol{R}_f$ and $\boldsymbol{R}_c$ represent matrices that are randomly sampled once and remain constant during the training process. d_s represents the sampled dimensions. When the dimension $d_f \times d_c$ exceeds 1024, a random multilinear mapping $T_{\odot}(\boldsymbol{f}, \boldsymbol{c})$ is employed; otherwise, $T_{\otimes}(\boldsymbol{f}, \boldsymbol{c})$ is used.

Assume that the data distribution of the source and target domains is represented as $P_s(\boldsymbol{x})$ and $P_t(\boldsymbol{x}')$, respectively, the loss function L_{domain} for the discriminator D is optimized during training according to the following objective equation:

$$\min_D \max_T L_{\text{domain}} = -E_{\boldsymbol{x}_i^s \sim P_s(\boldsymbol{x})} \log\left[D\left(T\left(\boldsymbol{f}_i^s, \boldsymbol{c}_i^s\right)\right)\right] - E_{\boldsymbol{x}_j^t \sim P_t(\boldsymbol{x}')} \log\left[1 - D\left(T\left(\boldsymbol{f}_j^t, \boldsymbol{c}_j^t\right)\right)\right]. \quad (7)$$

Here, $D(,)$ represents the discriminator's predicted probability that $\boldsymbol{x}$ is from the source domain, while $1 - D(,)$ represents the discriminator's predicted probability that $\boldsymbol{x}$ originates from the target domain. For optimization of discriminator D, the above discriminative error function is minimized, while on the feature extractor f_ω and the classifier C, L_{domain} is maximized. This action is taken to render the discriminator incapable of differentiating whether the input samples originate from the source or the target domain. As a result, the data distributions of these two domains become harmonized.

2) Domain Separation Branch Network: Unlike simultaneously learning features from multiple domains, using different filters to learn discriminative features from different domains. It can prevent the model from biasing toward the information of the source domain for more labeled samples. In the above domain fusion branch, data distribution alignment is achieved. In this study, the domain separation learning branch is introduced to be implemented simultaneously with domain fusion. In this branch, feature learning with deep network embedding in the DG mechanism is proposed. By activating different filters for each domain, this structured development enables separate learning of discriminative knowledge for each domain. The gate mechanism as mentioned by Liang et al. [36] allows for the use of different filters to control information learning, and the relationship between filters and categories is one-to-one or one-to-many. The objective of this study is to establish a one-to-one correspondence between domains and filters to learn discriminative knowledge from both domains simultaneously. Therefore, DG that activates and utilizes the corresponding filters only when the input sample comes from a certain domain is proposed to be embedded in the network design. Shallow networks typically achieve common feature learning for two domain data. As illustrated in Fig. 1, parallel to the domain fusion branch, the DG is integrated into the branch network of E_3 to learn domain discriminative knowledge. This domain separation branch comprises a residual block, a max-pooling layer, and a 3D-CNN. The difference between the 3-D residual block in E_3 and the shallow learning module is that the number of convolution kernels is 16. The DG is inserted before the residual block and the 3D-CNN, serving as a filter selector before the convolutional operation.

In practical usage, the correlation between domains and filters is controlled by a gate matrix $\boldsymbol{M} \in [0, 1]^{d \times ch}$, where each value is initialized randomly. Here, d signifies the count of domains and ch denotes the number of filters. In this

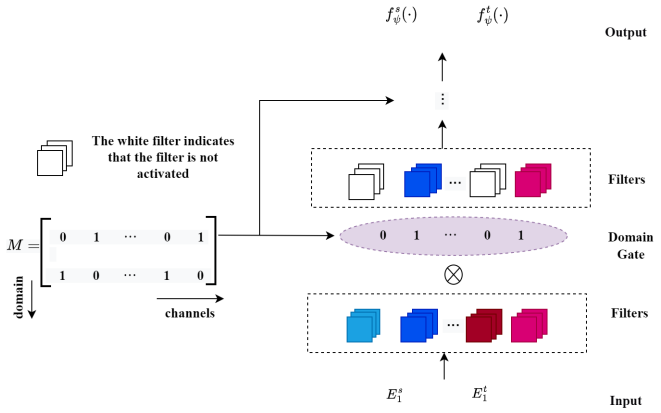

Fig. 3. Illustration of domain separation controlled by the DG matrix. The gating matrix functions in the gating network to achieve a one-to-one correspondence between domains and filters through selective activation of corresponding filters. This helps achieve domain separation and enables the network to independently learn discriminative knowledge from the source and target domains.

article, there are only two domains, so d is 2. $\boldsymbol{M}$ denotes the gate matrix for the source domain, and $1 - \boldsymbol{M}$ represents the gate matrix for the target domain. $\boldsymbol{M}_d^{ch} \in [0, 1]$ is the correlation between the dth domain and the chth filter, a higher value of which indicates a stronger correlation rendering the filter with a higher likelihood to be selected. However, this soft gate cannot effectively separate the two domains. For example, when $\boldsymbol{M}_d^{ch} = 0.5$, it is uncertain whether the filter is activated or not. Moreover, gate matrix data is discrete, and backpropagation does not have gradients for discrete variables. Therefore, in this article, Gumbel softmax [37] is introduced to generate hard gates, achieving a one-to-one correspondence between the domain and the filter. As shown in Fig. 3, the value of $\boldsymbol{M}_d^{ch}$ is either 0 or 1.0 signifies filter inactivation and 1 indicates filter activation.

Using the source-domain dataset as an instance, let f_ψ represent the overall learning function of shallow network E_1 and deep network E_1 and the deep network E_3, the support set feature representation after E_3 is represented as $f_\psi(\boldsymbol{x}_i)$, and the query set feature is represented as $f_\psi(\boldsymbol{x}_j)$. Combining $f_\psi(\boldsymbol{x}_i)$ and $f_\psi(\boldsymbol{x}_j)$ together as depicted in the following equation:

$$\boldsymbol{X}_o = \text{con}\left(f_\psi(\boldsymbol{x}_i), f_\psi\left(\boldsymbol{x}_j\right)\right) \quad (8)$$

where $o = 1, \ldots, N(K + W)$. Correspondingly, the joint sample label set is represented as Y_o. Finally, the feature $\boldsymbol{X}_o$ is fed into the global classifier G for classification. The loss function used for its optimization is the cross-entropy loss function. In each fragment $\boldsymbol{F}_s$ of the source-domain dataset, the class distribution of $\boldsymbol{X}_o$ is calculated as follows:

$$P_\psi(Y_o = k | \boldsymbol{X}_o \in \boldsymbol{F}_s) = \frac{\exp(-d(\boldsymbol{X}_o, \boldsymbol{n}_k))}{\sum_{k=1}^{N} \exp(-d(\boldsymbol{X}_o, \boldsymbol{n}_k))} \quad (9)$$

where d represents the Euclidean distance and $\boldsymbol{n}_k$ denotes the deep network output features of the kth class with gate integration in the fragment. Accordingly, the source-domain global loss for each fragment is calculated using the

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TABLE I
DETAILS OF ALL SIX HYPERSPECTRAL IMAGERY DATASETS USED

Datasets	Band Num.	Spectral Range(nm)	Spatial Resolution(m)	Image Size	Sensor	Class Num.
Chikusei	128	363-1018	2.5	2517×2335	Hyperspecr-VNIR-C	19
Indian Pines	200	400-2500	20	145×145	AVIRIS	16
Salinas	204	400-2500	3.7	512×217	AVIRIS	16
PaviaUniversity	103	430-860	1.3	610×340	ROSIS	9
WHU-Hi-LongKou	270	400-1000	0.463	550×400	Nano-Hyperspec	9
WHU-Hi-HanChuan	274	400-1000	0.109	1217×303	Nano-Hyperspec	16

following formula:

$$L^{s}_{\text{global}} = E_{\boldsymbol{F}_s}\left[-\sum_{(\boldsymbol{X},Y)\in\boldsymbol{F}_s}\log P_{\psi}(Y_o = k \mid \boldsymbol{X}_o)\right]. \tag{10}$$

Similar to the source domain, the global loss for each fragment in the target domain can be expressed as shown in the following equation:

$$L^{t}_{\text{global}} = E_{\boldsymbol{F}_t}\left[-\sum_{(\boldsymbol{X}',Y)\in\boldsymbol{F}_t}\log P_{\psi}\left(Y_o = k \mid \boldsymbol{X}'_o\right)\right]. \tag{11}$$

The proposed dual-branch domain adaptation FSL network simultaneously implements the domain separation and domain fusion branches to address the issue of diminishing differences in class distributions between the two domains due to domain distribution alignment. Finally, the computation of the total loss functions for FSL in the two domains, respectively, is given by the following equations:

$$L^{s}_{\text{total}} = L^{s}_{\text{global}} + L^{s}_{fsl} + L_{\text{domain}} \tag{12}$$

$$L^{t}_{\text{total}} = L^{t}_{\text{global}} + L^{t}_{fsl} + L_{\text{domain}}. \tag{13}$$

Therefore, the total loss function of the dual-branch domain adaptation FSL network is the sum of the aforementioned two parts, expressed as follows:

$$L_{\text{total}} = L^{s}_{\text{total}} + L^{t}_{\text{total}}. \tag{14}$$

Through optimization of (14), the proposed dual-branch FSL method effectively learns both transferable common knowledge and discriminative knowledge from two domains. During the testing phase, a restriction of labeled data from the target domain is applied. The test samples go through the shallow learning module which has been trained with the source domain data to achieve dimensionality reduction and spatial–spectral feature learning. Subsequently, deep network E_2 follows to realize higher-level semantic learning. Finally, a K-nearest neighbor (KNN) classifier is trained to achieve predictive classification of unlabeled target-domain data.

III. EXPERIMENTS AND RESULT

A. Datasets

To investigate the different performances in transfer learning from low-spatial-resolution images to high-spatial-resolution ones and the opposite process, six public hyperspectral datasets are utilized in our study. The first four are satellite/airborne datasets of Chikusei,[1] Indian Pines,[2] Salinas[3] and Pavia University,[4] which have relatively low spatial resolution. Additionally, two hyperspectral datasets of high spatial resolution are also used, which were captured by UAVs from Wuhan University, denoted as WHU-Hi-LongKou and WHU-Hi-HanChuan.[5] The detailed imaging parameters of those six datasets are provided in Table I. It needs to pay attention to the different spatial resolutions of these datasets. It can be seen that the first four datasets are all with spatial resolution over 1 m, and the lowest resolution even reaches 20 m. The Chikusei dataset acquired using the hyperspectral visible/near-infrared cameras (Hyperspec-VNIR-CIRIS) spectrometer has a spatial resolution of 2.5 m. The Indian Pines dataset was imaged using the airborne visible/infrared imaging spectrometer (AVIRIS) infrared imaging spectrometer with the worst spatial resolution of approximately 20 m. The Salinas dataset was also captured using AVIRIS but with a spatial resolution of 3.7 m. The Pavia University dataset was collected using the airborne hyperspectral sensor Reflective Optics System Imaging Spectrometer (ROSIS) with a spatial resolution of 1.3 m. Different from the above four well-known satellite/airborne datasets, the WHU-Hi-Longkou dataset, and the WU-Hi-HanChuan dataset were obtained using the Nano-Hyperspec imaging sensor mounted on the drone platform. Both these datasets have relatively high spatial resolution below 1 m. The former represents a simple agricultural area in Longkou, Hubei, China, on July 17, 2018, with an approximate resolution of 0.463 m. The latter one depicts a rural–urban fringe area of Hanchuan City in Hubei Province, China, on June 17, 2016, with the highest spatial resolution of 0.109 m. In addition, the six datasets have different numbers of spectral bands and all the numbers of bands are larger than 100. Therefore, it is reasonable to perform dimensionality reduction

[1] https://www.sal.t.u-tokyo.ac.jp/hyperdata/

[2] https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Indian_Pines

[3] https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Salinas

[4] https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_University_scene

[5] http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

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TABLE II
CLASS TYPES OF SIX DATASETS

Datasets	Class Types
Chikusei	(1) Water,(2) Bare soil(school),(3) Bare soil(park),(4) Bare soil(farmland),(5) Natural plants,(6) Weeds in farmland,(7) Forest,(8) Grass,(9) Rice field (grown),(10) Rice field(first stage),(11) Row crops, (12) Plastic house,(13) Manmade(non-dark),(14) Manmade(dark),(15) Manmade(blue), (16) Manmade(red),(17) Manmade grass,(18) Asphalt,(19) Paved ground
IndianPines	(1) Alfalfa,(2) Corn-notill, (3) Corn-mintill,(4) Corn,(5) Grass-pasture,(6) Grass-tree,(7) Grass-pasture -mowed,(8) Hay-windrowed,(9) Oats,(10) Soybean-notill,(11) Soybean-mintill,(12) Soybean-clean, (13) Wheat,(14) Woods,(15) Building-Grass-Trees-Drives,(16) Stone-Steel-Towers
Salinas	(1) Brocoli_green_weeds_1,(2) Brocoli_green_weeds_2,(3) Fallow, (4) Fallow_rough_plow,(5)Fallow_smooth, (6) Stubble,(7) Celery, (8) Grapes_untrained, (9) Soil_vinyard_develop, (10) Corn_senesced_green_weeds, (11) Lettuce_romaine_4wk,(12) Lettuce_romaine_5wk,(13) Lettuce_romaine_6wk,(14) Lettuce_romaine_7wk, (15) Vinyard_untrained,(16) Vinyard_vertical_trellis
PaviaUniversity	(1) Asphalt,(2) Meadows,(3) Gravel,(4) Trees,(5) Sheets,(6) Bare soil,(7) Bitumen,(8) Bricks,(9) Shadow
WHU-Hi-LongKou	(1) Corn,(2) Cotton,(3) Sesame,(4) Broad-leaf soybean,(5) Narrow-leaf soybean,(6) Rice,(7) Water, (8) Roads and houses,(9) Mixed weed
WHU-Hi-HanChuan	(1) Strawberry,(2) Cowpea,(3) Soybean,(4) Sorghum,(5) Water spinach,(6) Watermelon,(7) Greens, (8) Trees,(9) Grass,(10) Red roof,(11) Grey roof,(12) Plastic,(13) Bare soil,(14) Road, (15) Bright object,(16) Water

to unify both source and target data into 100 dimensions in the proposed network. Besides spectral and spatial resolution, another important difference between those datasets is their numbers of landscape class types which are summarized in Table II. In general, datasets that have more classes are preferred to be used as source data in transfer learning.

B. Comparison Algorithms

To evaluate the efficacy of the proposed approach, the traditional HSI classification method of support vector machines (SVMs) [38] and deep-learning methods including the latest cross-domain FSL methods mentioned in Section I are used for comparison. These deep-learning counterpart methods include 3D-CNN [39], DFSL + nearest neighbor (NN) [33], DFSL + SVM [33], DCFSL [28], and CTFSL [31].

SVM : Support vector machines utilize a kernel function to transform nonlinear input data from a low-dimensional to a high-dimensional linearly separable space. It is widely used for effective classification and regression analysis. However, SVM primarily focuses on spectral information while overlooking spatial information.

$3D-CNN$: 3D-CNN enables spectral–spatial feature extraction from HSIs. It provides a more comprehensive feature representation which is potential for classification accuracy improvement.

$DFSL+NN, DFSL+SVM$: These methods leverage the source-domain training data to learn a deep residual 3-D convolutional metric space capable of distinguishing different categories. Subsequently, a limited subset of labeled target-domain data is used to train NN or SVM classifiers within this metric space. Finally, the classifier is used for the classification of unlabeled data.

$DCFSL$: Building upon the foundation of DFSL, DCFSL addresses both domain adaptation and FSL within a unified framework. In this approach, after a metric space is learned through deep residual 3-D convolution. FSL is then performed in this metric space. Finally, a conditional adversarial network is utilized to harmonize domain data distributions and achieve domain adaptation.

$CTFSL$: It employs a CT network as a feature extractor to extract local–global features, and subsequently, a fully convolutional network (FCN) follows as the domain discriminator to achieve domain alignment utilizing adversarial loss.

C. Experimental Setup

The SVM and 3D-CNN methods involve training with merely a limited quantity of labeled target data. For cross-domain learning experiments with DFSL + NN, DFSL + SVM, DCFSL, and CTFSL, the training data consists of both source data and a limited set of labeled target data. To ensure fairness in the experiments, within the cross-domain setting, a total of 200 labeled samples per class are randomly extracted from the source data and utilized for training purposes. In all experiments, five labeled samples are randomly chosen from each category of the target data as the training set, while the rest data serves as the test set. To prevent the model from favoring the source domain with more labeled samples during training, random Gaussian noise is added to the five selected labeled target samples, resulting in an increased sample count of 200. During the training stage, every episode signifies an N-way-K-shot task. Here, K is set to 1 for the support set samples and Q is set to 19 for the query set samples. Additionally, a cube size of 9×9 is chosen for the input image, and optimization employs

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TABLE III
CLASSIFICATION RESULTS (%) ON DIFFERENT DATASETS WITH FIVE TRAINING SAMPLES PER CLASS (CHIKUSEI AS SOURCE DOMAIN)

Dataset	Indicator	Classification algorithms						
		SVM	3D-CNN	DFSL+NN	DFSL+SVM	DCFSL	CTFSL	Proposed Method
Indian Pines	OA(%)	47.15±1.38	53.42±0.18	65.38±2.98	65.60±2.91	67.11±1.67	69.69±1.32	**70.43±1.26**
	AA(%)	63.17±1.93	63.00±0.52	76.94±2.77	77.34±2.29	78.34±1.31	**80.83±1.17**	80.27±1.75
	Kappa(%)	41.03±1.75	46.69±0.48	61.04±2.12	61.16±2.38	62.75±0.97	65.62±1.29	**66.42±0.85**
Salinas	OA(%)	79.24±2.31	85.63±0.43	86.95±0.50	86.59±0.47	89.50±1.68	**92.01±1.10**	91.87±1.53
	AA(%)	89.79±1.95	92.66±1.31	93.61±1.07	93.50±0.29	94.31±1.27	**95.85±0.94**	95.45±1.42
	Kappa(%)	77.19±2.35	84.05±1.59	85.52±0.68	85.14±1.93	88.34±1.16	**91.11±1.12**	90.95±1.79
Pavia University	OA(%)	63.83±3.26	69.28±0.91	80.18±2.34	80.20±2.66	83.52±1.64	84.98±1.74	**85.50±1.33**
	AA(%)	69.68±1.50	72.72±1.19	80.55±2.67	80.54±2.87	84.09±1.42	**85.84±1.63**	85.09±1.52
	Kappa(%)	54.70±2.89	60.57±0.82	74.31±1.98	74.43±1.45	78.54±1.85	80.64±2.06	**81.01±0.79**
WHU-Hi-LongKou	OA(%)	82.01±0.93	86.13±0.81	86.36±0.57	85.90±0.73	91.37±0.55	90.40±1.08	**92.30±0.86**
	AA(%)	78.07±1.62	74.37±1.05	87.59±0.83	76.04±0.65	**90.71±0.49**	90.19±0.98	90.60±0.65
	Kappa(%)	77.31±1.85	82.07±1.16	82.79±0.44	81.97±0.84	88.89±0.87	87.69±1.11	**90.05±0.42**
WHU-Hi-HanChuan	OA(%)	46.31±1.68	55.52±1.57	57.54±1.33	57.60±1.75	62.70±0.80	66.46±0.89	**67.17±0.94**
	AA(%)	41.52±1.75	53.27±1.63	50.82±0.82	52.83±1.61	58.54±0.73	63.09±0.91	**63.23±0.81**
	Kappa(%)	38.92±1.01	49.91±1.39	51.91±1.17	51.95±1.25	57.66±0.86	61.80±0.95	**62.47±0.79**

the Adam optimizer. Model training spans 10 000 iterations, executed with a learning rate of 0.001.

The evaluation standards consist of overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). To minimize experimental variability, the experiments were replicated ten times, and the outcomes are presented as the respective average and standard deviation of three evaluation criteria across the ten iterations.

D. Experimental Results

To assess the performance in cross-transfer learning experiments, the low-spatial-resolution dataset of Chikusei and the high-spatial-resolution dataset of WHU-Hi-HanChuan are alternatively utilized as source-domain data to learn transferable knowledge. The reason for choosing Chikusei as the source domain is that among the four low-spatial-resolution hyperspectral datasets, the Chikusei dataset contains the most abundant landscape categories among all the datasets. Rich and diverse class types in the source-domain data enable the provision of extensive knowledge support for the transfer task. For the same concern, WHU-Hi-HanChuan is picked as the source domain from the two UAV high-spatial-resolution hyperspectral datasets.

Upon option of the Chikusei dataset as the source-domain data, the remaining five datasets are employed as target domains. Table III reports the classification results, respectively, on each dataset. The optimal classification accuracy achieved on each dataset is highlighted in bold. Clearly, all the domain-adaptive FSL methods (DFSL + NN, DFSL + SVM, DCFSL, CTFSL, and proposed method) outperform nondomain-adaptive methods (SVM, 3D-CNN). Among all the FSL methods, CTFSL is the most potential competitor of the proposed network. By designing an FSL method that aligns domains while preserving the independence of category distributions, our proposed approach ranks the first with the highest OA and Kappa metrics nearly on all the datasets except for Salinas. CTFSL only behaves slightly better than our approach in transfer learning on the Salinas dataset. Compared to the best counterpart of CTFSL, our proposed method shows different levels of improvement of transfer learning accuracy on the other five target datasets. This includes the Indian Pines and WHU-Hi-HanChuan datasets, which have a larger disparity between spatial resolutions of the source and target domain datasets. 0.74% improvement of OA and 0.8% gain of Kappa are observed on the Indian Pines dataset. The improvement amount on the WHU-Hi-HanChuan dataset is approaching. These two target datasets both have 16 classes and include many class types of crops similar to the Chikusei dataset. When using transfer learning from the Chikusei dataset to classification of the WHU-Hi-LongKou UAV hyperspectral data, the proposed approach is found the most improvement over CTFSL, 1.9 % improvement of OA and 2.36% of Kappa are observed. Datasets of Chikusei and WHU-Hi-LongKou also have similar class types, but the transferred knowledge is only used to classify nine classes. The results here demonstrate, compared to the counterparts, that our network enables more effective transfer learning for classes in common. Even though the Salinas dataset also has 16 classes, its landscape class types are mainly related to different weeds, and rarely are they similar to those in Chikusei data. After all, these results collectively underscore the efficacy of our approach in aligning domain distributions while preserving category independence and the feasibility of transferring knowledge from low- to high-spatial-resolution HSIs.

Figs. 4–8 visually showcase the classification maps of various approaches on different hyperspectral data. The visual observation results are in accordance with the comparative results in Table III. To visually highlight the classification results of these categories, red boxes are used to indicate areas with distinct classification performance in Figs. 4–8. Compared to other counterparts, our approach achieves more accurate classification results on categories misclassified by other methods, such as class 11 (Soybean-mintill) in the Indian Pines dataset, class 8 (Grapes_untrained) in the Salinas dataset, class 2 (Meadows) in the Pavia University dataset, class 4 (Broad-leaf soybean) in the WHU-Hi-LongKou dataset, and class 1 (Strawberry) in the WHU-Hi-HanChuan

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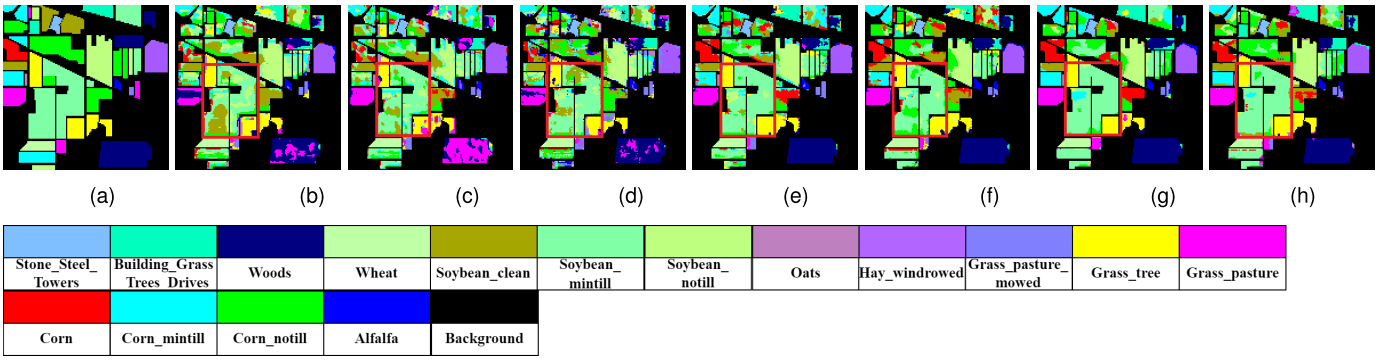

Fig. 4. Classification maps of Indian Pines HSI (Chikusei as source-domain data) achieved by various approaches with five training samples per class. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) DFSL + NN. (e) DFSL + SVM. (f) DCFSL. (g) CTFSL. (h) Proposed method.

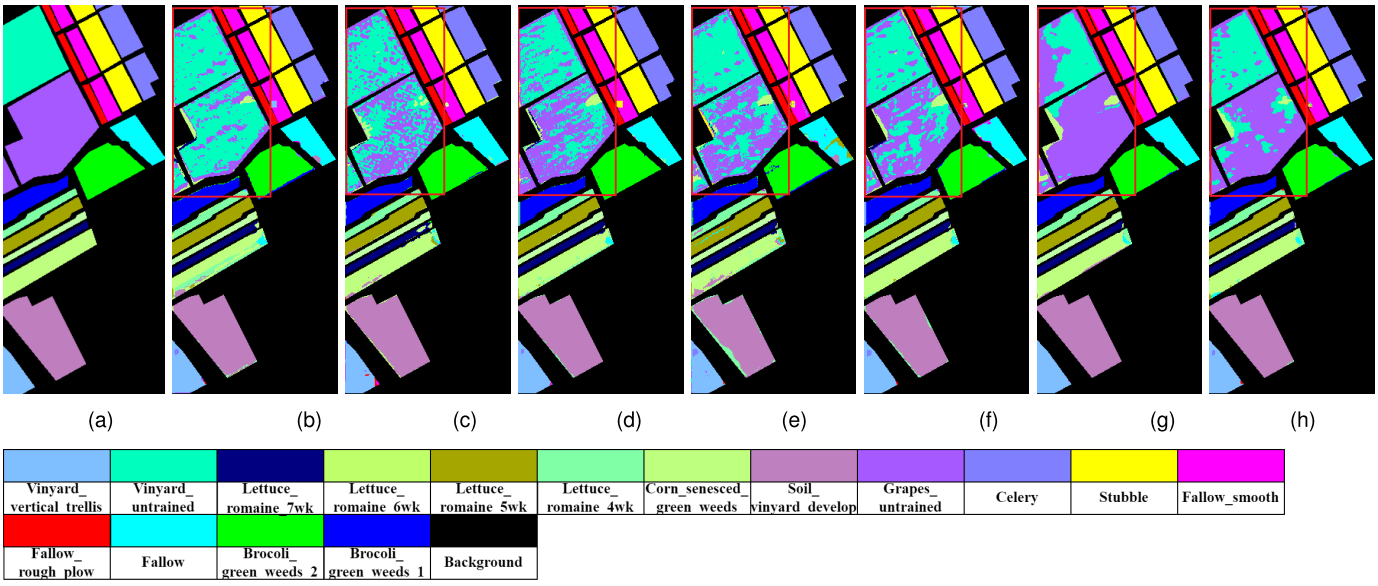

Fig. 5. Classification maps of Salinas HSI (Chikusei as source domain data) achieved by various approaches with five training samples per class. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) DFSL + NN. (e) DFSL + SVM. (f) DCFSL. (g) CTFSL. (h) Proposed method.

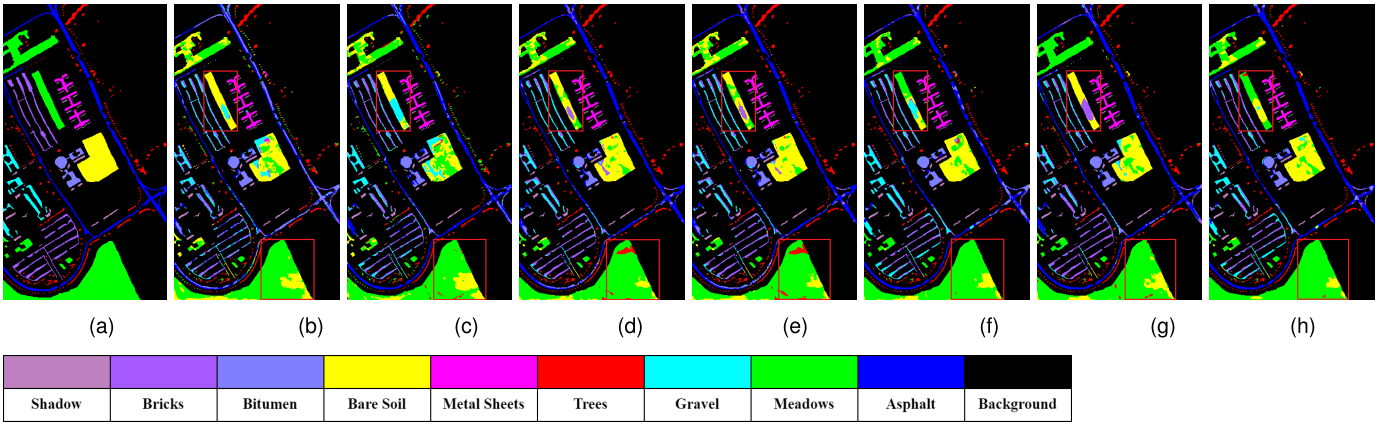

Fig. 6. Classification maps of Pavia University HSI (Chikusei as source-domain data) achieved by various approaches with five training samples per class. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) DFSL + NN. (e) DFSL + SVM. (f) DCFSL. (g) CTFSL. (h) Proposed method.

dataset. The above-mentioned categories mainly involve crops, which have significant differences from the categories in the source-domain Chikusei data. This demonstrates the importance of our method in learning discriminative knowledge from both the source and target domains. On the contrary, for class 8 (Hay-windrowed) in the Indian Pines dataset, class 16 (Vinyard_vertical_trellis) and class 9 (Soil_vinyard_develop) in the Salinas dataset, class 1 (Asphalt) and class 6 (Bare soil) in the Pavia University dataset, class 8 (Roads and houses) in the WHU-Hi-LongKou dataset, and class 10 (Red roof) in the WHU-Hi-HanChuan dataset, CTFSL achieves better classification results than ours. Those class types show relatively small differences compared to the categories in the source-domain Chikusei dataset. In this case, CTFSL utilizing

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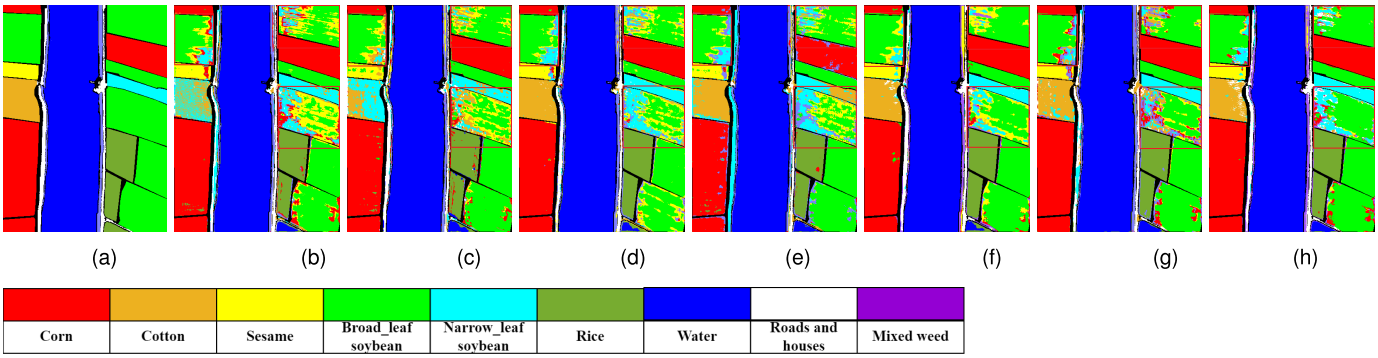

Fig. 7. Classification maps of WHU-Hi-LongKou HSI (Chikusei as source-domain data) achieved by various approaches with five training samples per class. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) DFSL + NN. (e) DFSL + SVM. (f) DCFSL. (g) CTFSL. (h) Proposed method.

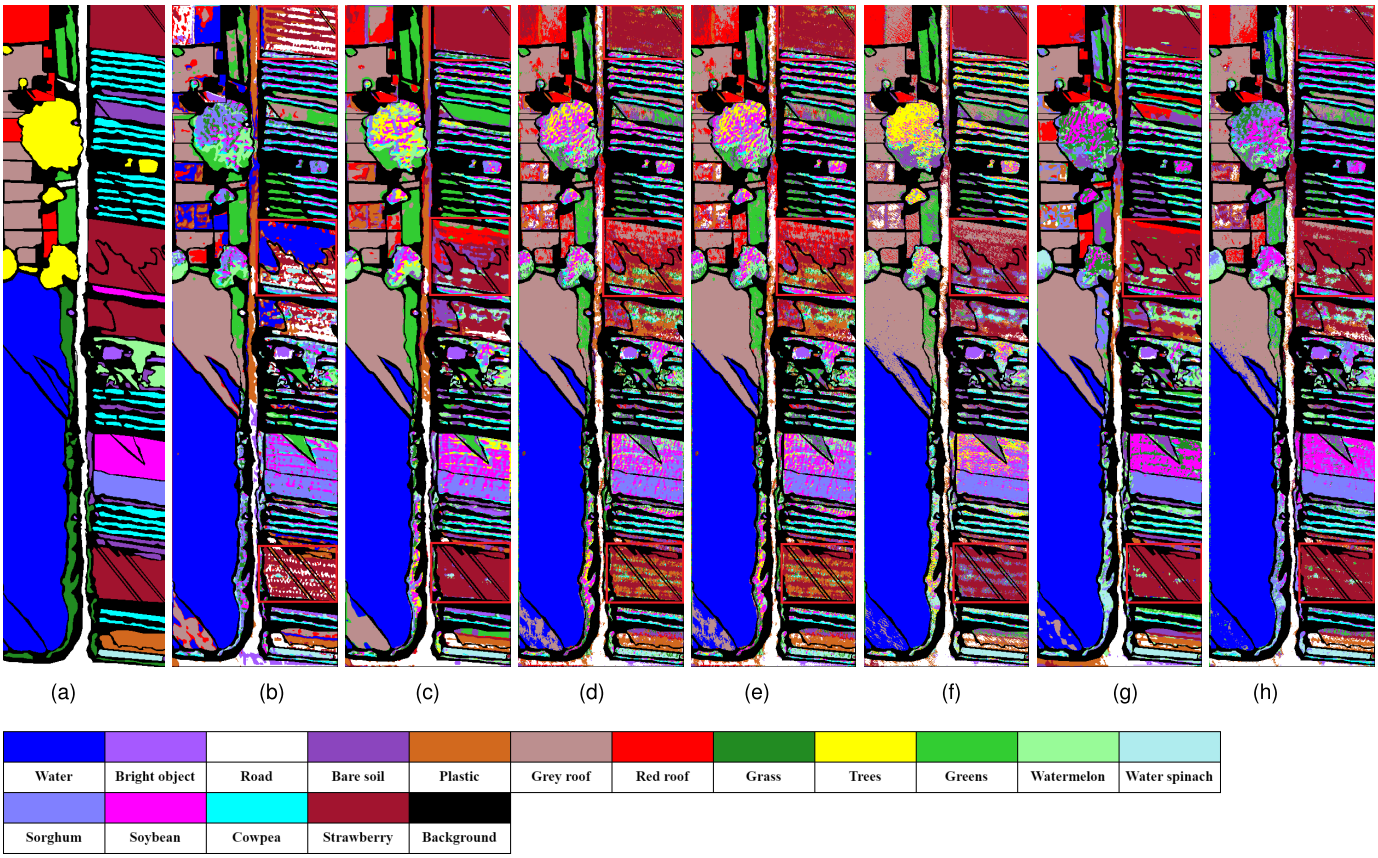

Fig. 8. Classification maps of WHU-Hi-HanChuan HSI (Chikusei as source-domain data) achieved by various approaches with five training samples per class. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) DFSL + NN. (e) DFSL + SVM. (f) DCFSL. (g) CTFSL. (h) Proposed Method.

a Transformer to learn both global and local information may play well in emphasizing the acquisition of domain-invariant knowledge.

To assess the efficacy of the proposed approach under contrary experimental conditions, in the following experiments, WHU-Hi-HanChuan is chosen as the source data for learning, while the other datasets are used as target data for prediction. A summary of all the outcomes can be found in Table IV. The comparison results of SVM and 3D-CNN are consistent with the results in Table III. Clearly, domain-adaptive FSL methods exhibit notably superior performance compared to nondomain-adaptive approaches. Compared to the second-best method of CTFSL, the hugest accuracy increase is also achieved on the dataset of Indian Pines which has the largest difference in spatial resolution with WHU-Hi-HanChuan. Around 1.8% gain of AA and Kappa is reported on the Indian Pines dataset. On the contrary, it is seen less than 0.3% accuracy improvement on the Chikusei dataset. This is because the source dataset of WHU-Hi-HanChuan has 16 classes, while the Chikusei dataset has 19 classes. There are new class types that cannot benefit from source data learning. For the results obtained on the rest two datasets, the accuracy improvement in terms of the two accuracy metrics is all over 1%. These results indicate the feasibility of transferring learning knowledge from high-spatial-resolution data to low-spatial-resolution HSIs.

Figs. 9–13 furthermore show the classification maps of various approaches on different hyperspectral data when using source-domain data of WHU-Hi-HanChuan. Despite it having higher spatial resolution and providing detailed spatial information, it has fewer categories compared to target-domain

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TABLE IV

CLASSIFICATION RESULTS (%) ON DIFFERENT DATASETS WITH FIVE TRAINING SAMPLES PER CLASS (WHU-HI-HANCHUAN AS THE SOURCE DOMAIN)

Dataset	Indicator	Classification algorithms						
		SVM	3D-CNN	DFSL+NN	DFSL+SVM	DCFSL	CTFSL	Proposed Method
Chikusei	OA(%)	83.64±1.47	87.29±0.77	89.34±1.54	89.31±1.28	92.49±0.94	92.96±1.05	**93.24±0.83**
	AA(%)	82.72±1.28	88.10±0.90	88.96±0.99	88.84±1.01	91.44±0.86	91.02±0.67	**91.87±0.98**
	Kappa(%)	81.29±1.32	85.43±0.86	87.76±1.30	87.73±1.17	91.36±1.08	91.90±1.01	**92.22±1.13**
Indian Pines	OA(%)	47.15±1.38	53.42±0.18	66.28±0.70	66.07±0.65	66.54±0.78	66.96±1.46	**68.70±1.55**
	AA(%)	63.17±1.93	63.00±0.52	77.62±0.85	77.55±1.03	**79.55±1.21**	79.42±1.50	78.24±1.67
	Kappa(%)	41.03±1.75	46.69±0.48	62.08±1.04	61.83±0.91	62.37±1.14	62.58±1.58	**64.30±1.40**
Salinas	OA(%)	79.24±2.31	85.63±0.43	89.13±0.82	89.55±1.09	90.60±1.37	92.07±0.82	**92.24±0.95**
	AA(%)	89.79±1.95	92.66±1.31	94.92±0.62	95.10±0.90	95.00±0.89	95.93±0.91	**96.34±1.07**
	Kappa(%)	77.19±2.35	84.05±1.59	87.95±0.94	88.41±0.77	89.55±0.73	91.17±1.04	**91.38±0.84**
Pavia University	OA(%)	63.83±3.26	69.28±0.91	80.56±0.83	80.77±0.75	83.39±0.94	83.53±1.71	**84.43±1.25**
	AA(%)	69.68±1.50	72.72±1.19	81.88±1.15	81.82±1.10	82.33±1.33	**83.87±1.57**	83.39±1.40
	Kappa(%)	54.70±2.89	60.57±0.82	75.13±1.00	75.26±1.06	78.29±0.80	78.73±1.88	**79.77±1.17**
WHU-Hi-LongKou	OA(%)	82.01±0.93	86.13±0.81	90.25±1.36	90.17±0.96	91.95±0.71	90.99±0.96	**93.31±1.15**
	AA(%)	78.07±1.62	74.37±1.05	87.36±1.28	87.52±1.58	90.88±0.44	90.72±0.82	**91.93±0.59**
	Kappa(%)	77.31±1.85	82.07±1.16	87.46±1.05	87.38±0.74	89.61±0.81	88.44±1.05	**91.34±0.78**

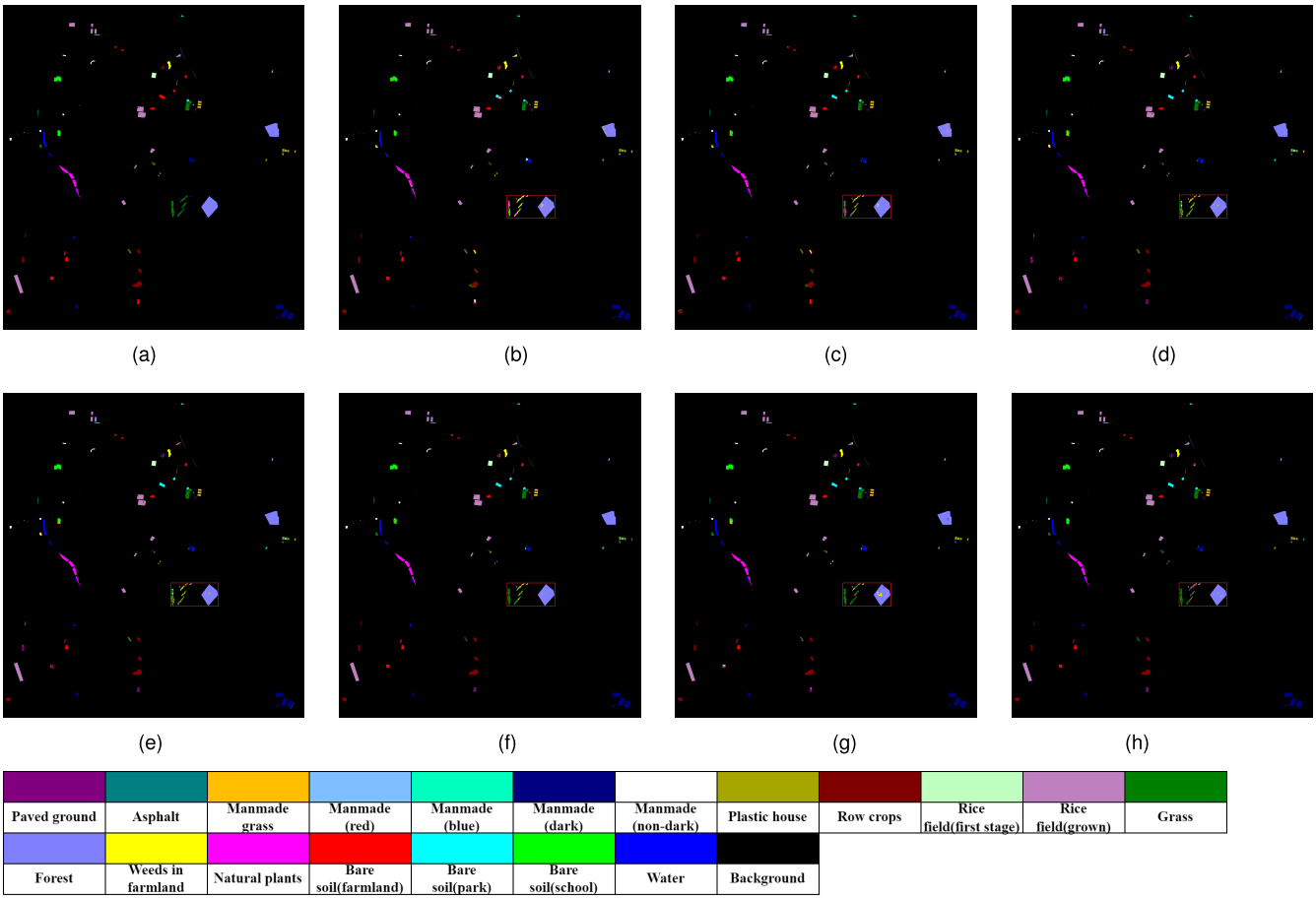

Fig. 9. Classification maps of Chikusei HSI (WHU-Hi-HanChuan as source-domain data) achieved by various approaches with five training samples per class. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) DFSL + NN. (e) DFSL + SVM. (f) DCFSL. (g) CTFSL. (h) Proposed method.

datasets of Chikusei, Indian Pines, and Salinas. Even under such conditions, the classification results of areas in the red boxes in those figures still show our method maintains superior classification performance on misclassified categories compared to other methods. For instance, in category 7 (Forest) and category 8 (Grass) in the Chikusei dataset, class 3 (Corn-mintill) and class 11 (Soybean-mintill) in the Indian Pines dataset, category 13 (Lettuce_romaine_6wk) in the Salinas dataset, as well as category 3 (Sesame) and category 4 (Broad-leaf soybean) in the WHU-Hi-LongKou dataset, the proposed approach generates results that are approaching the ground truth more closely. This implies the proposed method, which is capable of learning both domain-invariant knowledge and discriminative knowledge from both the source and target domains, is more robust in transfer learning when the source-domain data provides less transferable information.

Besides, by comparing figures in Tables III and IV, it can also be observed that, when Salinas and WHU-Hi-LongKou

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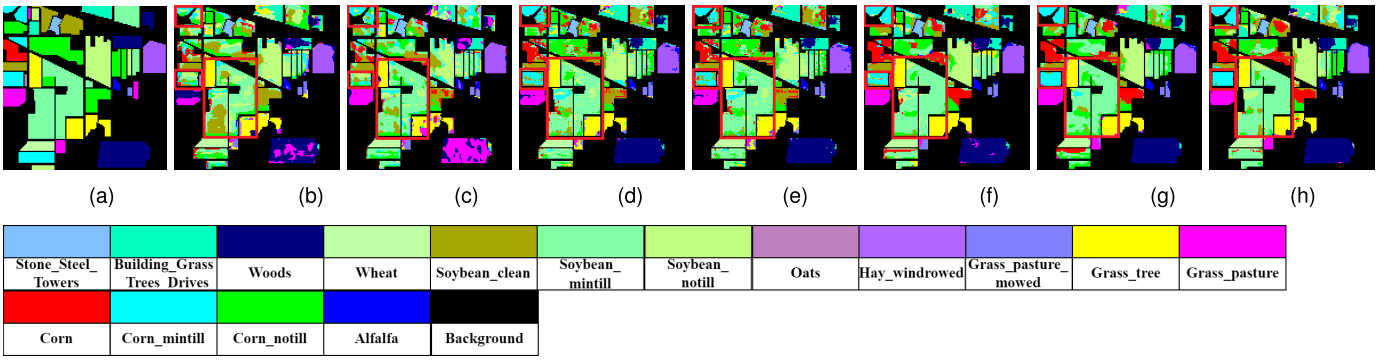

Fig. 10. Classification maps of Indian Pines HSI (WHU-Hi-HanChuan as source-domain data) achieved by various approaches with five training samples per class. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) DFSL + NN. (e) DFSL + SVM. (f) DCFSL. (g) CTFSL. (h) Proposed method.

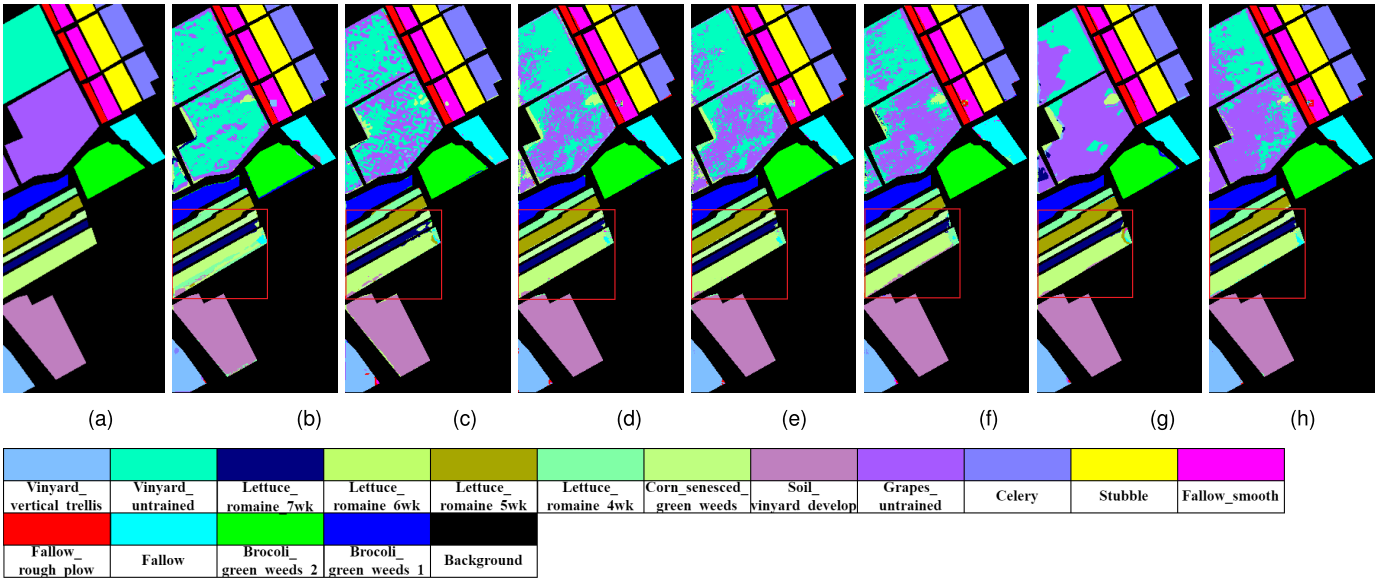

Fig. 11. Classification maps of Salinas HSI (WHU-Hi-HanChuan as source-domain data) achieved by various approaches with five training samples per class. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) DFSL+NN. (e) DFSL + SVM. (f) DCFSL. (g) CTFSL. (h) Proposed method.

are used as the target-domain dataset, the cross-domain classification performance achieved by learning from high-spatial-resolution data of WHU-Hi-HanChuan is superior to that learned from the low-spatial-resolution data of Chikusei. Specifically, the corresponding metrics obtained by the proposed method on the Salinas dataset in Table IV have furthermore improved based on the results in Table III (OA—0.37%, AA—0.89%, and Kappa—0.43%, respectively). On the WHU-Hi-LongKou, Table IV showed that the proposed method get over 1% improvement for the three accuracy metrics (OA—1.01%, AA—1.33%, and Kappa—1.29%, respectively) compared to that in Table III. These results seem to indicate that learning from hyperspectral data of high spatial resolution provides more abundant or potential information that can be transferred for enhancing the classification ability on the target-domain datasets.

However, opposite results were also observed when comparing the accuracy of the proposed method and CTFSL on Indian Pines and University of Pavia data. Comparisons of these figures in Tables III and IV show that the cross-domain classification performance achieved by learning from high-spatial-resolution data of WHU-Hi-HanChuan to these two target datasets is worse than that learned from low-spatial-resolution data of Chikusei. Different from the Chikusei dataset as the source dataset which has the most categories, the WHU-Hi-HanChuan dataset has fewer classes. It has 16 categories which are primarily related to crops such as Watermelon, Water spinach, Sorghum, and so on. Even though the source data of Indian Pines data also consists of 16 categories, those types are mainly related to vegetation and crops. There exists obvious diversity between crop types. As for the Pavia University dataset, it consists of nine classes like Asphalt, Bricks, Meadows, and more, which are significantly different from the categories in WHU-Hi-HanChuan. Those facts may result in a decrease in cross-domain classification performance. This discrepancy makes it challenging to transfer enough informative common knowledge from the source domain to the target domain.

To explore the influence of the number of labeled samples from the target data on classification performance, we conducted experiments using Chikusei as source-domain data. The quantity of labeled samples in the target domain for each class is systematically varied from 1 to 10 at increments of 1. These samples are selected randomly to form training sets for the

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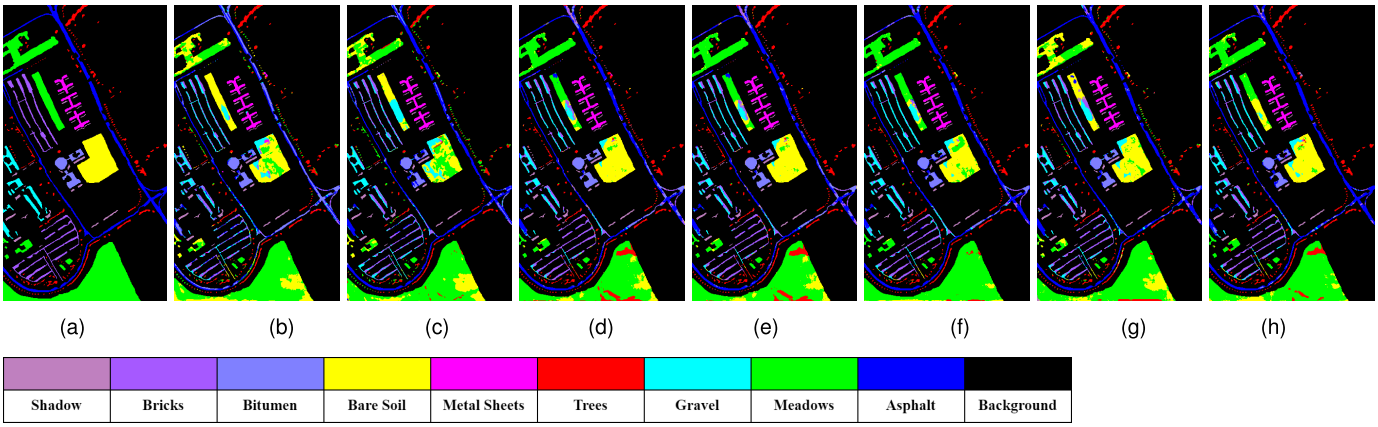

Fig. 12. Classification maps of Pavia University HSI (WHU-Hi-HanChuan as source-domain data) achieved by various approaches with five training samples per class. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) DFSL + NN. (e) DFSL + SVM. (f) DCFSL. (g) CTFSL. (h) Proposed method.

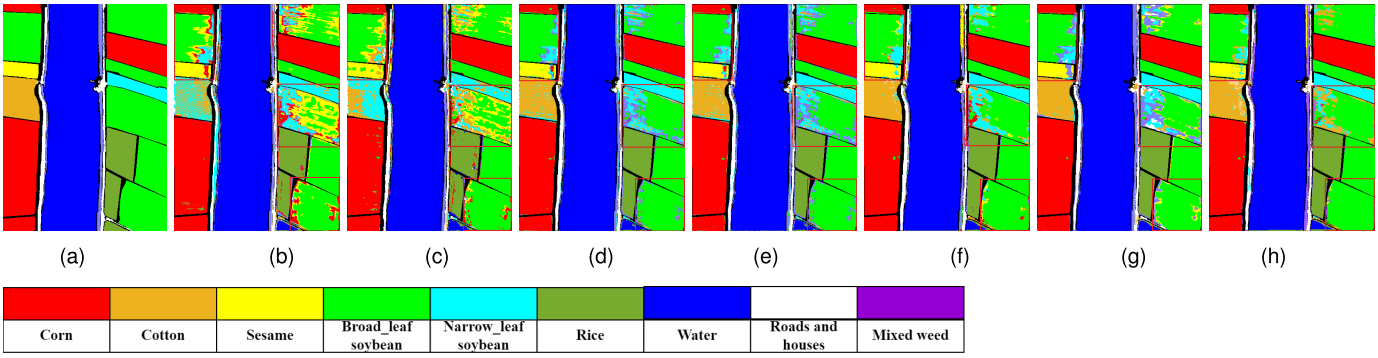

Fig. 13. Classification maps of WHU-Hi-LongKou HSI (WHU-Hi-HanChuan as source-domain data) achieved by various approaches with five training samples per class (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) DFSL + NN. (e) DFSL + SVM. (f) DCFSL. (g) CTFSL. (h) Proposed method.

TABLE V
COMPUTATION TIME AND PARAMETERS OF DIFFERENT METHODS ON DIFFERENT TARGET-DOMAIN DATASETS (CHIKUSEI AS SOURCE-DOMAIN DATA)

Dataset	Indicator	Classification algorithms						
		SVM	3D-CNN	DFSL+NN	DFSL+SVM	DCFSL	CTFSL	Proposed Method
Indian Pines	Training time(s)	0.29	82.38	1064.66	1024.27	1186.52	3277.70	1177.05
	Testing time(s)	0.46	1.73	1.60	1.08	1.64	6.23	1.98
	params	-	100844	34920	34920	4268994	10460243	4315526
Salinas	Training time(s)	1.13	93.21	1096.87	1178.91	1291.02	4134.60	1304.06
	Testing time(s)	2.71	9.75	4.39	5.32	4.92	5.37	4.74
	params	-	102892	34920	34920	4269394	10460643	4315926
Pavia University	Training time(s)	1.64	62.89	993.75	908.41	1010.20	2461.59	1079.57
	Testing time(s)	2.96	8.12	5.04	5.71	5.48	9.38	5.81
	params	-	50277	34920	34920	4259294	10447841	4307264
WHU-Hi-LongKou	Training time(s)	1.05	59.33	446.67	531.25	794.12	1370.65	835.24
	Testing time(s)	4.47	18.01	11.70	15.96	15.65	24.99	15.61
	params	-	136684	34920	34920	4275994	10464541	4323964
WHU-Hi-HanChuan	Training time(s)	3.57	70.21	693.62	746.73	1145.44	1829.66	1215.31
	Testing time(s)	7.38	30.23	17.03	20.65	20.48	32.97	20.57
	params	-	138732	34920	34920	4276394	10467643	4328872

target domain. The experiments are also repeated ten times across various labeled sample numbers, and the corresponding average and standard deviation of the representative metric of OA (three evaluation criteria) are computed as the accuracy results. Fig. 14 illustrates the OA curves of various techniques across different quantities of labeled data on five target-domain datasets. The plots in this figure clearly indicate the consistent ascending trend of classification accuracy of all methods as the count of labeled samples of each dataset increases. More careful observation can be found that before the number of training samples reaches 5, rapid growth is observed in most cases. When more training samples are supplied, the classification accuracy of each method increases gradually toward a stabilizing tendency. In comparison to counterpart approaches, the proposed method stands out in six cases with the most competitive accuracy and stable increasing trend. For other methods, the accuracy curves are witnessed with more times of variation. Therefore, it needs to be pointed out

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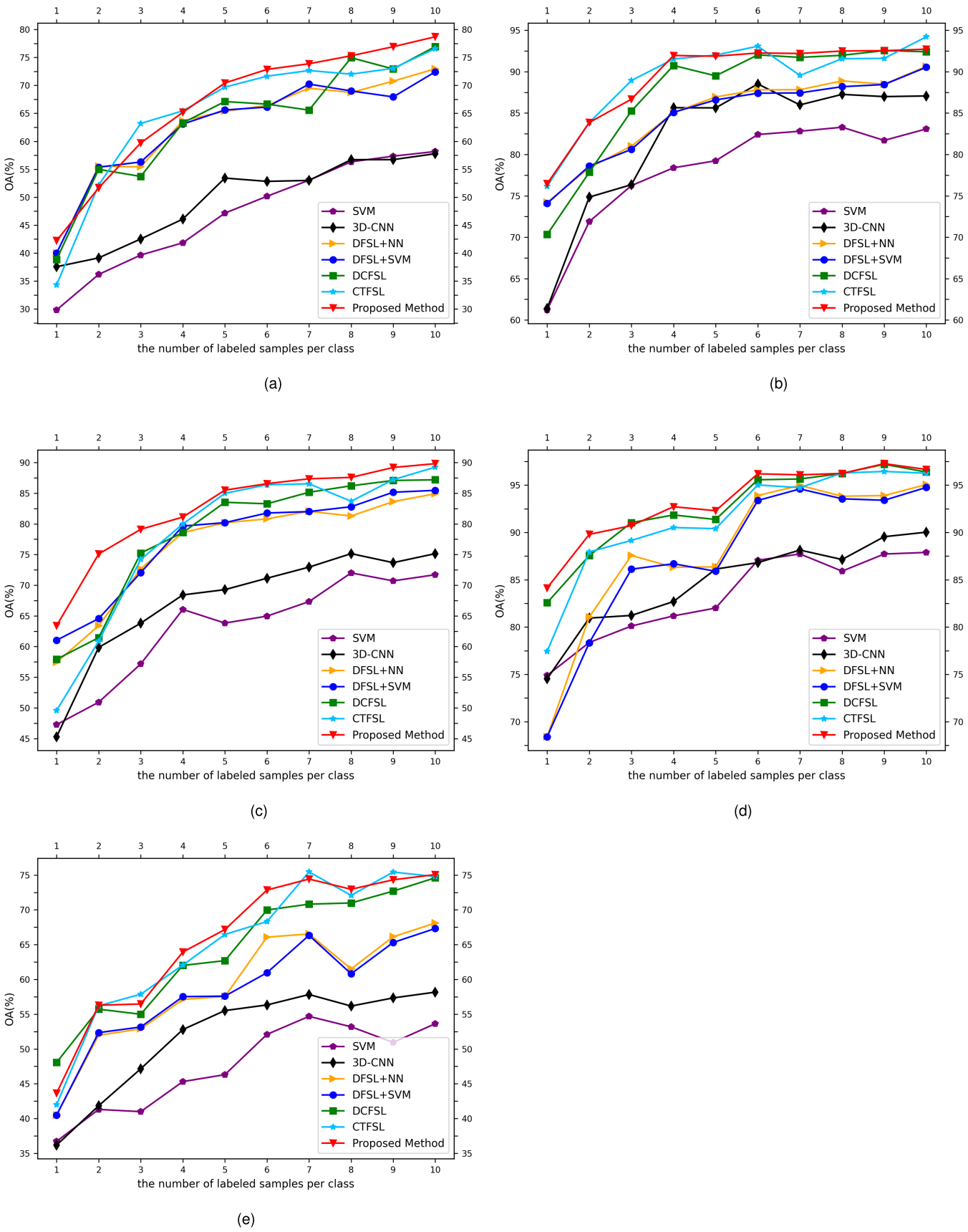

Fig. 14. Classification accuracy of various approaches using different numbers of labeled samples per class on six target datasets (Chikusei as source-domain data). (a) Indian Pines. (b) Salinas. (c) Pavia University. (d) WHU-Hi-LongKou. (e) WHU-Hi-HanChuan.

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that the experimental results in Section III-D are achieved by implementing each method using five labeled samples.

Besides the classification accuracy, the computational complexity of the method proposed is also provided in Table V. Taking the experiments using the Chikusei dataset as a source domain as an example, training time, test time, and the number of parameters needed to learn for each network are reported to intuitively demonstrate the computational efficiency of various methods. Training time of nondomain adaptation methods (SVM, 3D-CNN) indicates the time spent on training models on target-domain samples, while testing time refers to the time on predicting unlabeled samples in the target domain. On the other hand, domain adaptation methods (DFSL + NN, DFSL + SVM, DCFSL, CTFSL, and the proposed method) encompass the time training models in both the source and target domains, along with the transfer learning process. Testing time is the time predicting each unlabeled sample in the target domain. The figures in Table V indicate that compared to nondomain adaptation methods, domain adaptation networks take longer training time and have much more network parameters needed to be learned. In fact, they encounter higher computational costs associated with transfer learning. Among domain adaptation methods, our approach has more model parameters compared to DFSL + NN and DFSL + SVM. This is primarily due to the inclusion of a discriminator with five fully connected layers in the conditional adversarial network used during domain fusion. Additionally, the parameter count of our proposed network is similar to that of DCFSL. This shows that our proposed domain separation module learns discriminative features from the source and target domains very effectively without sacrificing model complexity. It is worth noting that the training time and parameter count of CTFSL are more than twice as much as ours, as CTFSL introduces a transformer for global feature extraction. While this enhances classification efficiency, it also significantly increases model complexity. Together with the previous classification accuracy results in Tables III and IV, the comprehensive results present that our approach behaves the best among the advanced domain adaptation methods after tradeoff between acceptable efficiency and precise classification.

IV. Conclusion

This article introduces an innovative neural network aimed at tackling the challenge of cross-domain FSL. A dual-branch domain adaptation network that combines domain fusion and domain separation is developed. The domain fusion branch aligns the data distributions of two domains using conditional adversarial networks, while the domain separation branch achieves class distribution independence through a gate mechanism. This approach effectively transmits knowledge garnered from the source domain to an entirely disparate target domain. The scope of knowledge transfer encompasses various scenarios, including the transfer from low-spatial-resolution HSIs to the classification task of high-spatial-resolution HSIs, and vice versa. Experimental results validate the superiority of this proposed technique, surpassing all existing cross-domain FSL methods in terms of classification capability. Therefore, this article provides an effective solution for addressing the cross-domain FSL problem in real-world scenarios. The potential of the proposed approach throws light on broad applications by extending this method to adapt to more practical application scenarios with different platforms and spectral sensors. It also needs to be noted that the potential for promising classification with the proposed method also relates to the spatial resolution difference between the two domain data, and the number and diversity of classes of source-domain data. Comparisons of the accuracies of the proposed method and CTFSL on Indian Pines and the University of Pavia data in the experiments present the limitations of our approach. When the number of categories in the source domain is fewer than that in the target domain or when the diversity of the source-domain dataset is relatively low compared to the target domain, the classification performance of the method does not demonstrate superiority over other methods. This is mainly because certain categories in the target domain cannot benefit from the learning in the source domain data. In other words, the learned knowledge from the source domain provides limited assistance in classifying the target domain. Therefore, based on this initial investigation conducted in this study, future work could explore more of those factors.

References

[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[2] X. Yang and Y. Yu, "Estimating soil salinity under various moisture conditions: An experimental study," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2525–2533, May 2017.

[3] M. Zhang, W. Li, X. Zhao, H. Liu, R. Tao, and Q. Du, "Morphological transformation and spatial-logical aggregation for tree species classification using hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501212.

[4] D. Zhu, B. Du, and L. Zhang, "Two-stream convolutional networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6907–6921, Aug. 2021.

[5] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[6] B. Tu, Q. Ren, C. Zhou, S. Chen, and W. He, "Feature extraction using multidimensional spectral regression whitening for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8326–8340, 2021.

[7] S. Jia, X. Deng, J. Zhu, M. Xu, J. Zhou, and X. Jia, "Collaborative representation-based multiscale superpixel fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7770–7784, Oct. 2019.

[8] S. Jia, Z. Zhan, and M. Xu, "Shearlet-based structure-aware filtering for hyperspectral and LiDAR data classification," *J. Remote Sens.*, vol. 2021, pp. 1–12, Jan. 2021.

[9] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[10] A. Vali, S. Comai, and M. Matteucci, "Deep learning for land use and land cover classification based on hyperspectral and multispectral Earth observation data: A review," *Remote Sens.*, vol. 12, no. 15, p. 2495, Aug. 2020.

[11] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[12] S. Mei, X. Chen, Y. Zhang, J. Li, and A. Plaza, "Accelerating convolutional neural network-based hyperspectral image classification by step activation quantization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502012.

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[13] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509612.

[14] S. Mei, J. Ji, Y. Geng, Z. Zhang, X. Li, and Q. Du, "Unsupervised spatial–spectral feature learning by 3D convolutional autoencoder for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6808–6820, Sep. 2019.

[15] H. Xu, W. He, L. Zhang, and H. Zhang, "Unsupervised spectral–spatial semantic feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526714.

[16] R. Fan, R. Feng, L. Wang, J. Yan, and X. Zhang, "Semi-MCNN: A semisupervised multi-CNN ensemble learning method for urban land cover classification using submeter HRRS images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4973–4987, 2020.

[17] S. Jia et al., "A semisupervised Siamese network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516417.

[18] X. Li, Z. Cao, L. Zhao, and J. Jiang, "ALPN: Active-Learning-Based prototypical network for few-shot hyperspectral imagery classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[19] H. L. Yang and M. M. Crawford, "Domain adaptation with preservation of manifold geometry for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 543–555, Feb. 2016.

[20] X. Ma, X. Mou, J. Wang, X. Liu, J. Geng, and H. Wang, "Cross-dataset hyperspectral image classification based on adversarial domain adaptation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4179–4190, May 2021.

[21] Y. Huang et al., "Two-branch attention adversarial domain adaptation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540813.

[22] Y. Huang et al., "Cross-scene wetland mapping on hyperspectral remote sensing images using adversarial domain adaptation network," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 37–54, Sep. 2023.

[23] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, pp. 2817–2830, Feb. 2021.

[24] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[25] C. Zhang, J. Yue, and Q. Qin, "Global prototypical network for few-shot hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4748–4759, 2020.

[26] Y. Wang, Q. Yao, J. T. Kwok, and L. M. Ni, "Generalizing from a few examples: A survey on few-shot learning," *ACM Comput. Surv.*, vol. 53, no. 3, pp. 1–34, May 2021.

[27] K. Gao, B. Liu, X. Yu, J. Qin, P. Zhang, and X. Tan, "Deep relation network for hyperspectral image few-shot classification," *Remote Sens.*, vol. 12, no. 6, p. 923, Mar. 2020.

[28] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[29] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 30, 2022, doi: 10.1109/TNNLS.2022.3185795.

[30] F. Xu, G. Zhang, C. Song, H. Wang, and S. Mei, "Multiscale and cross-level attention learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501615.

[31] Y. Peng, Y. Liu, B. Tu, and Y. Zhang, "Convolutional transformer-based few-shot learning for cross-domain hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1335–1349, 2023.

[32] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[33] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[34] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 4080–4090.

[35] M. Long, Z. Cao, J. Wang, and M. I. Jordan, "Conditional adversarial domain adaptation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 1640–1650.

[36] H. Liang et al., "Training interpretable convolutional neural networks by differentiating class-specific filters," in *Computer Vision—ECCV 2020*. Cham, Switzerland: Springer, 2020, pp. 622–638.

[37] E. Jang, S. Gu, and B. Poole, "Categorical reparameterization with gumbel-softmax," 2016, *arXiv:1611.01144*.

[38] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[39] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

Zhuowei Wang received the B.S. degree in computer science and technology from the China University of Geosciences, Wuhan, China, in 2007, and the M.S. and Ph.D. degrees in computer systems architecture from Wuhan University, Wuhan, in 2009 and 2012, respectively.

From 2019 to 2020, she worked as a Visiting Scholar with the Norwegian University of Science and Technology, Gjøvik, Norway. She is currently a Professor with the School of Computer Science and Technology, Guangdong University of Technology, Guangzhou, China. Her research interests focus on high-performance computing, low-power optimization, and distributed systems.

Shihui Zhao received the B.S. degree from Chuzhou University, Anhui, China, in 2015. She is currently pursuing the master's degree with the School of Computer Science and Technology, Guangdong University of Technology, Guangzhou, China.

Her research interests include hyperspectral image processing, computer vision, and deep learning.

Genping Zhao received the Ph.D. degree in information and telecommunications engineering from Harbin Engineering University, Harbin, China, in 2017.

From December 2013 to January 2015, she worked as a Visiting Ph.D. Student with the University of Western Australia, Perth, WA, Australia. From November 2018 to November 2019, she worked as a Post-Doctoral Fellow with the University of Alberta, Edmonton, AB, Canada. She is currently a Lecturer with the School of Computer Science and Technology, Guangdong University of Technology, Guangzhou, China. Her research interests focus on multisource remote-sensing data analysis and machine learning.

Xiaoyu Song received the Ph.D. degree from the University of Pisa, Pisa, Italy, in 1991.

From 1992 to 1998, he was a Faculty Member with the University of Montreal, Montreal, QC, Canada. He joined the Department of Electrical and Computer Engineering, Portland State University, Portland, OR, USA, in 1998, where he is currently a Professor. His research interests include formal methods, design automation, embedded computing systems, and emerging technologies.

Dr. Song was awarded an Intel Faculty Fellowship from 2000 to 2005. He was an Editor of IEEE TRANSACTIONS ON VLSI SYSTEMS and IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS.

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